

A Study of GPS Jamming and Anti-jamming

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Abstract—The weaknesses of GPS (Global Positioning System) is vulnerable to be jammed, so the study of jamming and anti-jamming of GPS has drawn more and more attention. In this paper, it analyzes the performance of GPS jamming and the anti-jamming of GPS receiver about blanket jamming. In addition, it establishes the simulation model based on MATLAB and gives the analysis of the performance of the anti-jamming about GPS receiver by adding the C/A code jamming and single-frequency jamming. Finally, based on the simulation, it analyzes the effect on GPS receiver by adding C/A code jamming and single-frequency jamming; and it achieves the analysis of the anti-jamming of GPS after the bit error decision, which proves that GPS have some ability of anti-noise and anti-jamming.

Index Terms—GPS, C/A code jamming, single-frequency jamming, anti-jamming

I. INTRODUCTION

GPS is a new generation of navigation and positioning system established by the U.S. Department of Defense to meet the military need of high precision navigation and positioning of the sea, the land and air facilities. It has a global, round-the-clock, continuous three-dimensional precision navigation and positioning capabilities [1]. However, GPS receiving is especially vulnerable to be jammed. At present, the applications of GPS on the military and civilian in China have been increasing in scope and volume, and therefore the study of GPS jamming technology should also be accelerated. The study not only covers GPS jamming technology to make the other side's advanced equipment fail, but also focuses on the advanced technology of GPS anti-jamming.

In this paper, it mainly achieves the study of GPS jamming and anti-jamming about blanket jamming; and it achieves the simulation of the C/A code jamming and single-frequency jamming to the GPS receiver by MATLAB. Then it analyzes the anti-jamming of GPS receiver according to the simulation results.

II. THE TECHNOLOGY OF GPS JAMMING

GPS receivers receive a very weak signal from the satellite, so the jamming of the receiver is a major GPS jamming means. From a technical point of view, the jamming of the receiver can be divided into two kinds: one as blanket jamming, and the other as deception jamming.

Blanket jamming is fired by strong jamming signals, blocking receiver, so that GPS signals can not be normal amplified and detected, which finally resulted in GPS receiver's reduction or completely lost of ability to work [2]. Blanket jamming can be divided into code aiming jamming and obstruction jamming.

It mainly focuses on obstruction jamming in the paper. Obstruction jamming includes single-frequency jamming and C/A code jamming. Obstruction jamming get into the power of GPS signal carrier frequency and bandwidth and jamming signals in a certain region may disrupt receiving the signal.

C/A code jamming is a broadband jamming. It jams the receiver to receive satellite signals through the weak relevance between different C/A codes. Single-frequency jamming is a narrow-band jamming. It is achieved spread-spectrum with pseudo-code in the receiver and becomes the widen-band jamming singles. There are little jamming signals can be though the narrow-band filter which can play the role of jamming [3].

III. GPS ANTI-JAMMING ANALYSIS

According to Shannon theory, in the spread spectrum communication, the bandwidth can be increased to reduce the signal-to-noise band in the same channel capacity. Even if the useful signals are covered by the noise, the system can work properly [4] [5].

In spread spectrum communication, the spreading gain (G_p) is defined the signal-to-noise ratio of the de-spread devices' output and the signal-to-noise ratio of receiver input, which represents the ability of suppressing the jamming signal which is input in receiver and enhancing the signal at the same time via spread-spectrum processing. The bigger G_p is the stronger anti-jamming ability. So as for the ability of anti-jamming spread-spectrum systems, it is necessary to analyze the processing gain. GPS signal is modulated by a pseudo-code before sending out; and the rate of the pseudo-code is much higher than the initial data, which can extend the data bandwidth wider. The GPS signal of spread spectrum is sent out after being modulated by carrier again and power amplification, as the following (1). So GPS is virtually spread spectrum communications and G_p is a key indicator of anti-jamming ability to measure of GPS as a DSSS system [6].

$$s(t) = Ad(t)PN(t) \cdot \exp(j(\omega_0 t + \varphi)) \quad (1)$$

In where, $d(t)$ is the data signal, $PN(t)$ is the pseudo-code and ω_0 is the frequency of the carrier.

GPS uses two different Gold spreading codes, namely C/A code and P code with the rate of 1.023×10^6 bit/s and 10.23×10^6 bit/s respectively. P code is exclusive for military and does not open to the civil use. In addition, the navigation message data rate is 50 bit/s. According to the spreading gain formula, the C/A code and P code of spreading gain can be calculated [7].

For C/A code, there is

$$G_{C/A} = \frac{1.023 \times 10^6}{50} = 20460 = 43dB \quad (2)$$

For P code, there is

$$G_P = \frac{10.23 \times 10^6}{50} = 204600 = 53dB \quad (3)$$

According to ICD-GPS-200D (GPS interface control document), it uses the standard linear polarization antenna to receive the GPS signal. The smallest receiver power of C/A code in L1 channel is 160 dBw, the P code is 163 dBw, while internal receiver noise is about 130 dBw. Therefore, the noise level higher is about 30 dB more than the signal. As the GPS uses the spread spectrum communication to increase the signal bandwidth and reduce the signal noise ratio. So even if the signal strength is lower than the noise, GPS receivers can still work properly. According to (2) and (3), the signal modulated by C/A code, processed through the relevant, can be around 13 dB higher than the noise, and the signal modulated by P code about 23 dB higher. Therefore, GPS itself has a certain ability of anti-jamming and anti-noise.

IV. THE STRUCTURE OF EXPERIMENTAL PLATFORM

In this paper, it mainly studies the downlink of GPS satellite communications. The system simulation model is Fig. 1. Based on Fig. 1, it simulates the C/A code jamming and single-frequency jamming on GPS receiver in this paper.

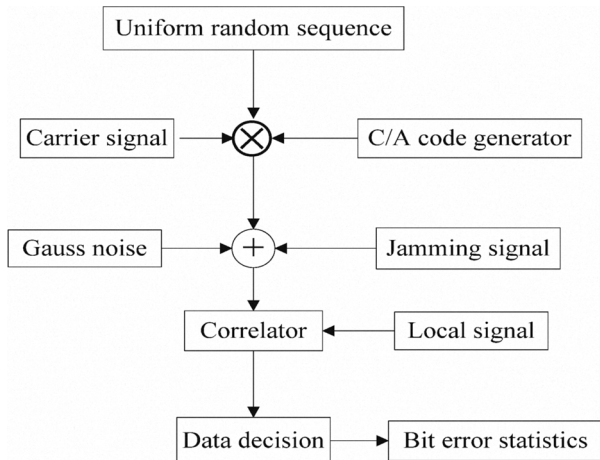


Figure 1. The simulation structure of the system

Seen from Fig. 1, the data is modulated by C/A code to achieve spread spectrum, and then gains the GPS satellite signals after BPSK (Binary Phase Shift Keying) modulation. GPS receivers receive GPS satellite signals and at the same time receive the Gauss noise and jamming signals from the environment. The GPS satellite signals mixed Gauss noise and jamming get into the receiver.

$$s_1(t) = Ad(t)PN(t) \cdot \exp(j(\omega_0 t + \varphi) + J(t) + n(t) \quad (4)$$

In where, $J(t)$ is the jamming signal and $n(t)$ is the gauss noise. ω_0 is the frequency of the carrier and φ is the inital phase of carrier.

Then the data code can be got after despreading and demodulation via local signals.

$$r(t) = s_1(t) \cdot PN(t - \tau) \cdot \exp(j(w_0 t + \varphi_1)) \quad (5)$$

In where, $PN(t - \tau)$ is the local pseudo-code and φ_1 is the inital phase of local carrier.

The bit error rate of the GPS system can be outputed after data decision and bit error statistics finally.

V. EXPERIMENT AND ANALYSIS

A. The Simulation of GPS Jamming

1) *the simulation of autocorrelation and cross correlation of the C/A code*: Because the spread spectrum codes of the GPS is essential to the process of signal regulation, GPS C/A has good property of autocorrelation and weak cross correlation. In order to test a weak signal, we need to make the auto peak higher than its cross peak with some stronger signal. It is very clear that if the code is perpendicular, the cross correlation value should be 0. But the gold code is not totally perpendicular, so the cross correlation value is not 0, and usually it should be a small number. The following Fig. 2 is the results of the simulation of the C/A code autocorrelation and cross correlation. In Fig. 2, the abscissa is the length of the data and the ordinate is the correlation value.

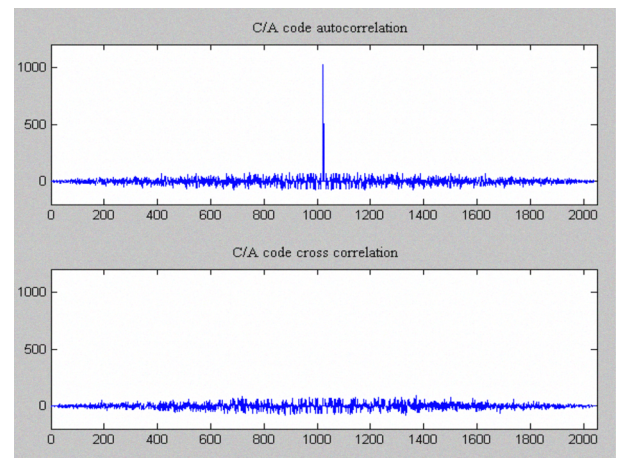


Figure 2. Results of the auto-correlation and cross-correlation of the C/A code

2) *the simulation of C/A code jamming and signal-frequency jamming*: Due to the little cross correlation between the C/A codes, the weak correlation of the C/A codes which the receiver receive from different satellites will jam the useful GPS signals. C/A code jamming is the jamming between different satellite signals received by the GPS receiver, seen from Fig. 3.

Single-frequency jamming usually causes some single-pinnacle noise, and if it masses near the carrier frequency of the spread spectrum signals, that will be very destructive, seen from Fig. 4.

In the Fig. 3 and Fig. 4, the top half shows the situation of the correlation value after data demodulated which happens before add jamming singles, and the correlation peak is relatively obvious; the bottom half is the situation of correlation value after data demodulated which happens after adding jamming, and the correlation peak is not very obvious and the effect of the jamming is relatively obvious. Comparing Fig. 3 and Fig. 4, it is shown that C/A code jamming more to the receiver than single-frequency jamming with the same jamming-to-signal ratio.

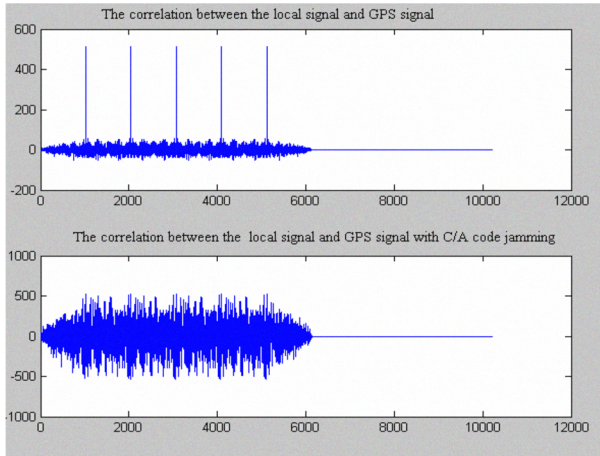


Figure 3. The simulation result of C/A code jamming

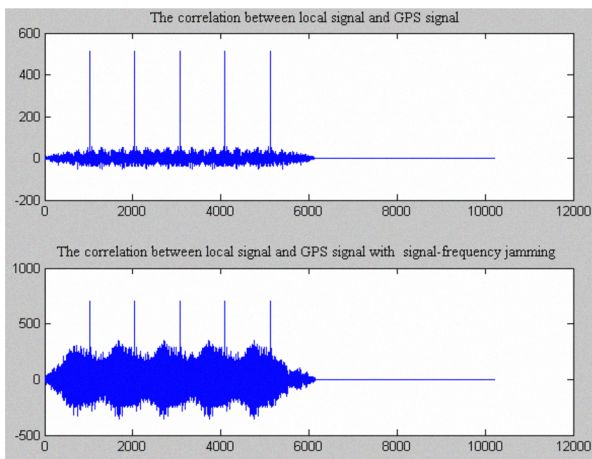


Figure 4. The simulation result of single-frequency jamming

B. The Analysis of GPS Anti-jamming

Before we jam to the GPS receiver, the bit error rate of the receiver is 0, seen from Fig. 5; after adding the Gauss noise, the variance of Gauss noise (W) is 0.63, the signal-to-noise is -1 dB, the signal has been totally swamped in the noise and the bit error rate linger near the point of 0, seen from Fig. 6. So, it explains that GPS system owns a very strong ability of anti-noise. In Fig. 5 and Fig. 6, the abscissa shows the cycle point, and the ordinate is the bit error rate about the receiver.

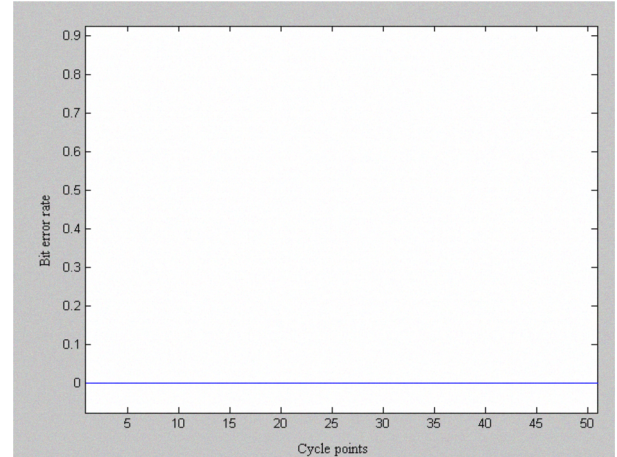


Figure 5. Bit error rate without noise

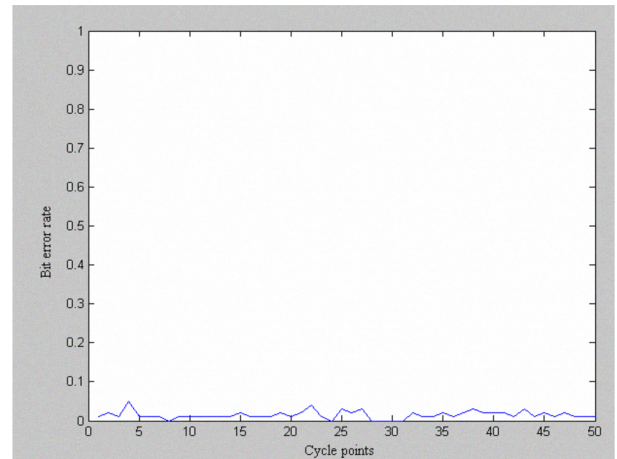


Figure 6. Bit error rate with gauss noise

Adding Gauss noise and jamming at the same time, bit error code rate is shown by Fig. 7 and Fig. 8.

The abscissa is the jamming-to-signal ratio and the ordinate is the bit error rate in Fig. 7 and Fig. 8. The variance of Gauss noise (W) is 0.63; signal-to-noise is -1 dB. And with the increase of the jamming-to-signal ratio, the bit error rate begins to rise. After adding into the C/A code jamming, jamming-to-signal ratio is about 21.6 dB and this time the bit error rate is 100%. When adding the single-frequency jamming, the jamming-to-signal ratio is 24.6 dB; the bit error rate reaches 100%. Comparing Fig. 7 and Fig. 8, with the same jamming-to-signal ratio which shows the C/A code jamming has higher bit error code rate, and the signal-frequency jamming has a larger number, when the bit error rate is 100%. So, the effect of C/A code jamming is better than the single-frequency jamming,

and it proves that GPS has better ability of anti single-frequency jamming.

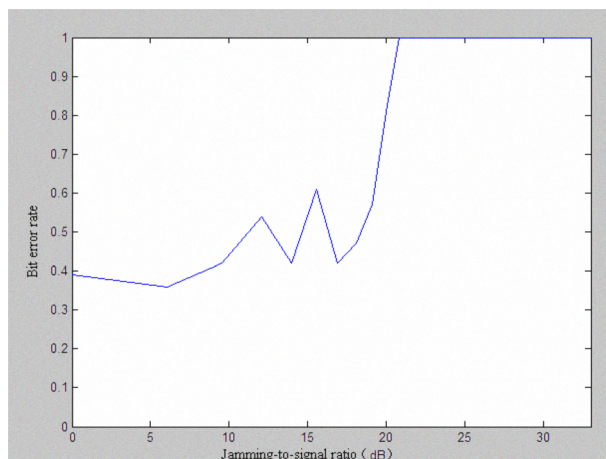


Figure 7. Bit error rate with adding C/A code jamming and gauss noise

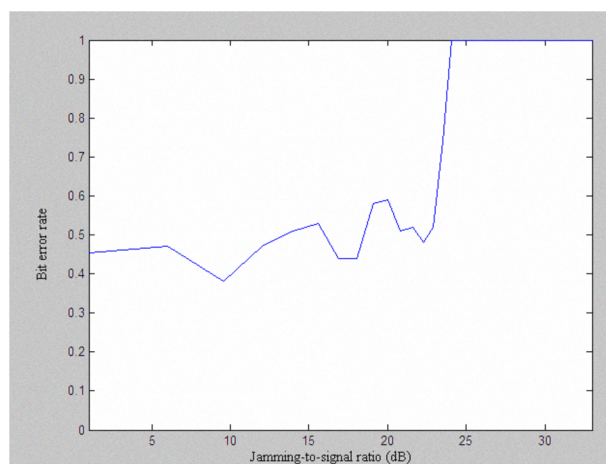


Figure 8. Bit error code rate with adding single-frequency jamming and gauss noise

VI. CONCLUSION

In the paper, it has simply introduced the GPS jamming technology and done some analysis about the anti-jamming ability of the GPS. In addition, it also achieves the simulation of C/A jamming and single-frequency jamming by building system simulation model. It can be seen from those experiments that no matter it is C/A jamming or single-frequency jamming, the bit error rate will raise with the increase of jamming-to-signal ratio; and when the jamming-to-signal ratio reaches some point, the system's bit error rate will suddenly rocket up and reach 1 which shows that system can not work properly because the limit of the jamming has been exceeded. Also, the effect of C/A code jamming to the GPS receiver is better than single frequency jamming, which proves the GPS system has better ability of anti-narrow-band jamming. The experiments and analysis mentioned above will be great help to the study of GPS anti-jamming technology in future.

ACKNOWLEDGMENT

The authors wish to thank all members of the GPS team for their hard work. This work was supported by the Scientific Research Foundation with Jiangxi Provincial Department of Education in China (Grant No. GJJ08243).

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